



NVM Express[®]

Base Specification

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Please send comments to info@nvmexpress.org

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Performance may be calculated using parameters returned as part of the SMART / Health Information log. Specifically, the number of Read or Write commands, the amount of data read or written, and the amount of controller busy time enables both I/Os per second and bandwidth to be calculated.

The log page returned is defined in Figure 210.

Figure 210: SMART / Health Information Log Page

Bytes	Description																		
00	Critical Warning (CW): This field indicates critical warnings for the state of the controller. Each bit corresponds to a critical warning type; multiple bits may be set to '1'. If a bit is cleared to '0', then that critical warning does not apply. Critical warnings may result in an asynchronous event notification to the host. Bits in this field represent the state at the time the Get Log Page command is processed and may not reflect the state at the time a related asynchronous event notification, if any, occurs or occurred. <table border="1"> <thead> <tr> <th>Bits</th><th>Description</th></tr> </thead> <tbody> <tr> <td>7</td><td>Reserved</td></tr> <tr> <td>6</td><td>Indeterminate Personality State (IPS): If this bit is set to '1', then a requested change to the settings of a personality did not complete successfully and the controller was not able to revert to the previous personality. As a result, the settings of a personality in the NVM subsystem are in an indeterminate state (refer to section 8.1.6.1).</td></tr> <tr> <td>5</td><td>Persistent Memory Region Read-Only (PMRRO): If this bit is set to '1', then the Persistent Memory Region has become read-only or unreliable (refer to section 8.2.4).</td></tr> <tr> <td>4</td><td>Volatile Memory Backup Failed (VMBF): If this bit is set to '1', then the volatile memory backup device has failed. This field is only valid if the controller has a volatile memory backup solution.</td></tr> <tr> <td>3</td><td>All Media Read-Only (AMRO): If this bit is set to '1', then all of the media has been placed in read only mode. The controller shall not set this bit to '1' if the read-only condition on the media is a result of a change in the write protection state of a namespace (refer to section 8.1.17.1).</td></tr> <tr> <td>2</td><td>NVM Subsystem Degraded Reliability (NDR): If this bit is set to '1', then the NVM subsystem reliability has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.</td></tr> <tr> <td>1</td><td>Temperature Threshold Condition (TTC): If this bit is set to '1', then a temperature is: a) greater than or equal to an over temperature threshold; or b) less than or equal to an under temperature threshold, (refer to section 5.2.26.1.3).</td></tr> <tr> <td>0</td><td>Available Spare Capacity Below Threshold (ASCBT): If this bit is set to '1', then the available spare capacity has fallen below the threshold.</td></tr> </tbody> </table>	Bits	Description	7	Reserved	6	Indeterminate Personality State (IPS): If this bit is set to '1', then a requested change to the settings of a personality did not complete successfully and the controller was not able to revert to the previous personality. As a result, the settings of a personality in the NVM subsystem are in an indeterminate state (refer to section 8.1.6.1).	5	Persistent Memory Region Read-Only (PMRRO): If this bit is set to '1', then the Persistent Memory Region has become read-only or unreliable (refer to section 8.2.4).	4	Volatile Memory Backup Failed (VMBF): If this bit is set to '1', then the volatile memory backup device has failed. This field is only valid if the controller has a volatile memory backup solution.	3	All Media Read-Only (AMRO): If this bit is set to '1', then all of the media has been placed in read only mode. The controller shall not set this bit to '1' if the read-only condition on the media is a result of a change in the write protection state of a namespace (refer to section 8.1.17.1).	2	NVM Subsystem Degraded Reliability (NDR): If this bit is set to '1', then the NVM subsystem reliability has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.	1	Temperature Threshold Condition (TTC): If this bit is set to '1', then a temperature is: a) greater than or equal to an over temperature threshold; or b) less than or equal to an under temperature threshold, (refer to section 5.2.26.1.3).	0	Available Spare Capacity Below Threshold (ASCBT): If this bit is set to '1', then the available spare capacity has fallen below the threshold.
Bits	Description																		
7	Reserved																		
6	Indeterminate Personality State (IPS): If this bit is set to '1', then a requested change to the settings of a personality did not complete successfully and the controller was not able to revert to the previous personality. As a result, the settings of a personality in the NVM subsystem are in an indeterminate state (refer to section 8.1.6.1).																		
5	Persistent Memory Region Read-Only (PMRRO): If this bit is set to '1', then the Persistent Memory Region has become read-only or unreliable (refer to section 8.2.4).																		
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2	NVM Subsystem Degraded Reliability (NDR): If this bit is set to '1', then the NVM subsystem reliability has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.																		
1	Temperature Threshold Condition (TTC): If this bit is set to '1', then a temperature is: a) greater than or equal to an over temperature threshold; or b) less than or equal to an under temperature threshold, (refer to section 5.2.26.1.3).																		
0	Available Spare Capacity Below Threshold (ASCBT): If this bit is set to '1', then the available spare capacity has fallen below the threshold.																		
02:01	Composite Temperature (CTEMP): Contains a value corresponding to a temperature in Kelvins that represents the current composite temperature of the controller and namespace(s) associated with that controller. The manner in which this value is computed is implementation specific and may not represent the actual temperature of any physical point in the NVM subsystem. The value of this field may be used to trigger an asynchronous event (refer to section 5.2.26.1.3). Warning and critical overheating composite temperature threshold values are reported by the WCTEMP and CCTEMP fields in the Identify Controller data structure in Figure 328.																		
03	Available Spare (AVSP): Contains a normalized percentage (0% to 100%) of the remaining spare capacity available.																		
04	Available Spare Threshold (AVSPT): When the Available Spare falls below the threshold indicated in this field, an asynchronous event completion may occur. The value is indicated as a normalized percentage (0% to 100%). The values 101 to 255 are reserved.																		

Figure 210: SMART / Health Information Log Page

Bytes	Description												
05	Percentage Used (PUSED): Contains a vendor specific estimate of the percentage of NVM subsystem life used based on the actual usage and the manufacturer's prediction of NVM life. A value of 100 indicates that the estimated endurance of the NVM in the NVM subsystem has been consumed, but may not indicate an NVM subsystem failure. The value is allowed to exceed 100. Percentages greater than 254 shall be represented as 255. This value shall be updated once per power-on hour (when the controller is not in a sleep state). Refer to the JEDEC JESD218B-02 standard for SSD device life and endurance measurement techniques.												
06	Endurance Group Critical Warning Summary (EGCWS): This field indicates critical warnings for the state of Endurance Groups. Each bit corresponds to a critical warning type, multiple bits may be set to '1'. If a bit is cleared to '0', then that critical warning does not apply to any Endurance Group. Critical warnings may result in an asynchronous event notification to the host. Bits in this field represent the current associated state and are not persistent. If a bit is set to '1' in one or more Endurance Groups, then the corresponding bit shall be set to '1' in this field. <table border="1"> <thead> <tr> <th>Bits</th><th>Description</th></tr> </thead> <tbody> <tr> <td>7:4</td><td>Reserved</td></tr> <tr> <td>3</td><td>Endurance Group Read-Only (EGRO): If this bit is set to '1', then the namespaces in one or more Endurance Groups have been placed in read only mode not as a result of a change in the write protection state of a namespace (refer to section 8.1.17.1).</td></tr> <tr> <td>2</td><td>Endurance Group Degraded Reliability (EGDR): If this bit is set to '1', then the reliability of one or more Endurance Groups has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.</td></tr> <tr> <td>1</td><td>Reserved</td></tr> <tr> <td>0</td><td>Endurance Group Available Spare Capacity Below Threshold (EGASCBT): If this bit is set to '1', then the available spare capacity of one or more Endurance Groups has fallen below the threshold.</td></tr> </tbody> </table>	Bits	Description	7:4	Reserved	3	Endurance Group Read-Only (EGRO): If this bit is set to '1', then the namespaces in one or more Endurance Groups have been placed in read only mode not as a result of a change in the write protection state of a namespace (refer to section 8.1.17.1).	2	Endurance Group Degraded Reliability (EGDR): If this bit is set to '1', then the reliability of one or more Endurance Groups has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.	1	Reserved	0	Endurance Group Available Spare Capacity Below Threshold (EGASCBT): If this bit is set to '1', then the available spare capacity of one or more Endurance Groups has fallen below the threshold.
Bits	Description												
7:4	Reserved												
3	Endurance Group Read-Only (EGRO): If this bit is set to '1', then the namespaces in one or more Endurance Groups have been placed in read only mode not as a result of a change in the write protection state of a namespace (refer to section 8.1.17.1).												
2	Endurance Group Degraded Reliability (EGDR): If this bit is set to '1', then the reliability of one or more Endurance Groups has been degraded due to significant media related errors or any internal error that degrades NVM subsystem reliability.												
1	Reserved												
0	Endurance Group Available Spare Capacity Below Threshold (EGASCBT): If this bit is set to '1', then the available spare capacity of one or more Endurance Groups has fallen below the threshold.												
31:07	Reserved												
47:32	Data Units Read (DUR): Contains the number of 512 byte data units the host has read from the controller as part of processing a SMART Data Units Read Command; this value does not include metadata. This value is reported in thousands (i.e., a value of 1 corresponds to 1,000 units of 512 bytes read) and is rounded up (e.g., one indicates that the number of 512 byte data units read is from 1 to 1,000, three indicates that the number of 512 byte data units read is from 2,001 to 3,000). Refer to the applicable NVM Express I/O Command Set specification for the list of SMART Data Units Read Commands that affect this field. A value of 0h in this field indicates that the number of SMART Data Units Read is not reported.												
63:48	Data Units Written (DUW): Contains the number of 512 byte data units the host has written to the controller as part of processing a User Data Out Command; this value does not include metadata. This value is reported in thousands (i.e., a value of 1 corresponds to 1,000 units of 512 bytes written) and is rounded up (e.g., one indicates that the number of 512 byte data units written is from 1 to 1,000, three indicates that the number of 512 byte data units written is from 2,001 to 3,000). Refer to the applicable NVM Express I/O Command Set specification for the list of User Data Out Commands that affect this field. A value of 0h in this field indicates that the number of Data Units Written is not reported.												
79:64	Host Read Commands (HRC): Contains the number of SMART Host Read Commands completed by the controller. Refer to the applicable NVM Express I/O Command Set specification for the list of SMART Host Read Commands that affect this field.												
95:80	Host Write Commands (HWC): Contains the number of User Data Out Commands completed by the controller. Refer to the applicable NVM Express I/O Command Set specification for the list of User Data Out Commands that affect this field.												

Figure 210: SMART / Health Information Log Page

Bytes	Description
111:96	Controller Busy Time (CBT): Contains the amount of time the controller is busy with I/O commands. The controller is busy when there is a command outstanding to an I/O Queue (specifically, a command was issued via an I/O Submission Queue Tail doorbell write and the corresponding completion queue entry has not been posted yet to the associated I/O Completion Queue). This value is reported in minutes.
127:112	Power Cycles (PWRC): Contains the number of power cycles.
143:128	Power On Hours (POH): Contains the number of power-on hours. This may not include time that the controller was powered and in a non-operational power state.
159:144	Unexpected Power Losses (UPL): Contains a count of the number of unexpected power losses. This count shall be incremented if, and only if, main power is lost when: a) the controller does not report it is ready to be powered off (i.e., the CSTS.SHST field is not set to 10b); or b) media is not in the shutdown state (refer to Figure 85) because an Admin command that accesses media as defined by Figure 84 was processed via the out-of-band mechanism with the Ignore Shutdown bit set to '1' (refer to the NVM Express Management Interface Specification) while shutdown processing was reported as in progress or was reported as complete (i.e., the CSTS.SHST field was set to 01b or set to 10b). Note that this field was previously named Unsafe Shutdowns. If power is lost when the CSTS.SHST field is not set to 10b, then subsequent controller initialization (i.e., the amount of time from when the CC.EN bit transitions from '0' to '1' until the CSTS.RDY bit transitions from '0' to '1') may take longer for any NVM subsystem, and data corruption may occur for any NVM subsystem that is not protected against power loss.
175:160	Media and Data Integrity Errors (MDIE): Contains the number of occurrences where the controller detected an unrecovered data integrity error. Errors such as uncorrectable ECC, CRC checksum failure, or LBA tag mismatch are included in this field. Errors introduced as a result of a Write Uncorrectable command (refer to the NVM Express NVM Command Set Specification) may or may not be included in this field.
191:176	Number of Error Information Log Entries (NEILE): Contains the number of Error Information Log Entries over the life of the controller.
195:192	Warning Composite Temperature Time (WCTT): If the Temperature Threshold Hysteresis Attributes (TMPTHHA) field cleared to 0h (refer to Figure 328), then this field contains the amount of time in minutes that the controller is operational and the Composite Temperature is greater than or equal to the Warning Composite Temperature Threshold (WCTEMP) field and less than the Critical Composite Temperature Threshold (CCTEMP) field in the Identify Controller data structure in Figure 328. If the Temperature Threshold Maximum Hysteresis (TMPTHMH) and Temperature Threshold Hysteresis (TMPTHH) fields are non-zero, then hysteresis time is included in the time specified in this field (refer to section 5.2.26.1.3.1). If the value of the WCTEMP or CCTEMP field is 0h, then this field is always cleared to 0h regardless of the Composite Temperature value.
199:196	Critical Composite Temperature Time (CCTT): Contains the amount of time in minutes that the controller is operational and the Composite Temperature is greater than or equal to the Critical Composite Temperature Threshold (CCTEMP) field in the Identify Controller data structure in Figure 328. If the value of the CCTEMP field is 0h, then this field is always cleared to 0h regardless of the Composite Temperature value.
201:200	Temperature Sensor 1 (TSEN1): Contains the current temperature reported by temperature sensor 1. This field is defined by Figure 211.
203:202	Temperature Sensor 2 (TSEN2): Contains the current temperature reported by temperature sensor 2. This field is defined by Figure 211.
205:204	Temperature Sensor 3 (TSEN3): Contains the current temperature reported by temperature sensor 3. This field is defined by Figure 211.
207:206	Temperature Sensor 4 (TSEN4): Contains the current temperature reported by temperature sensor 4. This field is defined by Figure 211.

Figure 210: SMART / Health Information Log Page

Bytes	Description
209:208	Temperature Sensor 5 (TSEN5): Contains the current temperature reported by temperature sensor 5. This field is defined by Figure 211.
211:210	Temperature Sensor 6 (TSEN6): Contains the current temperature reported by temperature sensor 6. This field is defined by Figure 211.
213:212	Temperature Sensor 7 (TSEN7): Contains the current temperature reported by temperature sensor 7. This field is defined by Figure 211.
215:214	Temperature Sensor 8 (TSEN8): Contains the current temperature reported by temperature sensor 8. This field is defined by Figure 211.
219:216	Thermal Management Temperature 1 Transition Count (TMT1TC): Contains the number of times the controller transitioned to lower power active power states or performed vendor specific thermal management actions while minimizing the impact on performance in order to attempt to reduce the Composite Temperature because of the host controlled thermal management feature (refer to section 8.1.18.5) (i.e., the Composite Temperature rose above the Thermal Management Temperature 1). This counter shall not wrap once the value FFFFFFFFh is reached. A value of 0h, indicates that this transition has never occurred or this field is not implemented.
223:220	Thermal Management Temperature 2 Transition Count (TMT2TC): Contains the number of times the controller transitioned to lower power active power states or performed vendor specific thermal management actions regardless of the impact on performance (e.g., heavy throttling) in order to attempt to reduce the Composite Temperature because of the host controlled thermal management feature (refer to section 8.1.18.5) (i.e., the Composite Temperature rose above the Thermal Management Temperature 2). This counter shall not wrap once the value FFFFFFFFh is reached. A value of 0h, indicates that this transition has never occurred or this field is not implemented.
227:224	Total Time For Thermal Management Temperature 1 (TTTMT1): Contains the number of seconds that the controller had transitioned to lower power active power states or performed vendor specific thermal management actions while minimizing the impact on performance in order to attempt to reduce the Composite Temperature because of the host controlled thermal management feature (refer to section 8.1.18.5). This counter shall not wrap once the value FFFFFFFFh is reached. A value of 0h, indicates that this transition has never occurred or this field is not implemented.
231:228	Total Time For Thermal Management Temperature 2 (TTTMT2): Contains the number of seconds that the controller had transitioned to lower power active power states or performed vendor specific thermal management actions regardless of the impact on performance (e.g., heavy throttling) in order to attempt to reduce the Composite Temperature because of the host controlled thermal management feature (refer to section 8.1.18.5). This counter shall not wrap once the value FFFFFFFFh is reached. A value of 0h, indicates that this transition has never occurred or this field is not implemented.
239:232	Operational Lifetime Energy Consumed (OLEC): This field contains the cumulative operational energy consumed by the NVM subsystem in watt-hours calculated from all interval power measurements (i.e., average of power measurement samples over one second, refer to section 8.1.19) collected from the time of manufacture to the point that this log page is read. This value is rounded up (e.g., two indicates the number of watt-hours consumed is greater than 1 and less than or equal to 2, three indicates the number of watt-hours consumed is greater than 2 and less than or equal to 3). This field shall not wrap once the value FFFFFFFF_FFFFFFFFh is reached. A value of 0h in this field indicates that the cumulative operational energy consumed by the NVM subsystem is not reported.

Figure 210: SMART / Health Information Log Page

Bytes	Description																						
243:240	Interval Power Measurement (IPM): This field contains the interval power measurement (i.e., average of power measurement samples over one second, refer to section 8.1.19) of the indicated power measurement type (i.e., the PMT field) over the most recent one second interval at the time of processing the Get Log Page command to read this log page. The power in Watts is equal to the value in the Interval Power Measurement Value field multiplied by the scale indicated in the Interval Power Measurement Scale field. A value of 0h in this field indicates that the interval power measurement is not reported.																						
	<table><tr><th>Bits</th><th>Description</th></tr><tr><td>31:24</td><td>Reserved</td></tr><tr><td>23:20</td><td>Power Measurement Type (PMT): This field contains the power measurement type for the Interval Power Measurement Value field as defined in Figure 679.</td></tr><tr><td>19:18</td><td>Reserved</td></tr><tr><td>17:16</td><td>Interval Power Measurement Scale (IPMS): This field contains the scale for the Interval Power Measurement Value field.<table><tr><th>Value</th><th>Definition</th></tr><tr><td>00b</td><td>Not reported</td></tr><tr><td>01b</td><td>0.0001 W</td></tr><tr><td>10b</td><td>0.01 W</td></tr><tr><td>11b</td><td>Reserved</td></tr></table></td></tr><tr><td>15:00</td><td>Interval Power Measurement Value (IPMV): This field contains the value of the interval power measurement. If the Interval Power Measurement Scale field is cleared to 00b, then this field shall be cleared to 0h.</td></tr></table>	Bits	Description	31:24	Reserved	23:20	Power Measurement Type (PMT): This field contains the power measurement type for the Interval Power Measurement Value field as defined in Figure 679.	19:18	Reserved	17:16	Interval Power Measurement Scale (IPMS): This field contains the scale for the Interval Power Measurement Value field. <table><tr><th>Value</th><th>Definition</th></tr><tr><td>00b</td><td>Not reported</td></tr><tr><td>01b</td><td>0.0001 W</td></tr><tr><td>10b</td><td>0.01 W</td></tr><tr><td>11b</td><td>Reserved</td></tr></table>	Value	Definition	00b	Not reported	01b	0.0001 W	10b	0.01 W	11b	Reserved	15:00	Interval Power Measurement Value (IPMV): This field contains the value of the interval power measurement. If the Interval Power Measurement Scale field is cleared to 00b, then this field shall be cleared to 0h.
	Bits	Description																					
	31:24	Reserved																					
	23:20	Power Measurement Type (PMT): This field contains the power measurement type for the Interval Power Measurement Value field as defined in Figure 679.																					
	19:18	Reserved																					
	17:16	Interval Power Measurement Scale (IPMS): This field contains the scale for the Interval Power Measurement Value field. <table><tr><th>Value</th><th>Definition</th></tr><tr><td>00b</td><td>Not reported</td></tr><tr><td>01b</td><td>0.0001 W</td></tr><tr><td>10b</td><td>0.01 W</td></tr><tr><td>11b</td><td>Reserved</td></tr></table>	Value	Definition	00b	Not reported	01b	0.0001 W	10b	0.01 W	11b	Reserved											
Value	Definition																						
00b	Not reported																						
01b	0.0001 W																						
10b	0.01 W																						
11b	Reserved																						
15:00	Interval Power Measurement Value (IPMV): This field contains the value of the interval power measurement. If the Interval Power Measurement Scale field is cleared to 00b, then this field shall be cleared to 0h.																						
511:244	Reserved																						

Figure 211: Temperature Sensor Data Structure

Bits	Description
15:00	Temperature Sensor Temperature (TST): Contains the current temperature in Kelvins reported by the temperature sensor. The physical point in the NVM subsystem whose temperature is reported by the temperature sensor and the temperature accuracy is implementation specific. An implementation that does not implement the temperature sensor reports a value of 0h. The temperature reported by a temperature sensor may be used to trigger an asynchronous event (refer to section 5.2.26.1.3).

5.2.12.1.4 Firmware Slot Information (Log Page Identifier 03h)

This log page is used to describe the firmware revision stored in each firmware slot supported. The firmware revision is indicated as an ASCII string. The log page also indicates the active slot number. The log page returned is defined in Figure 212.

Figure 212: Firmware Slot Information Log Page

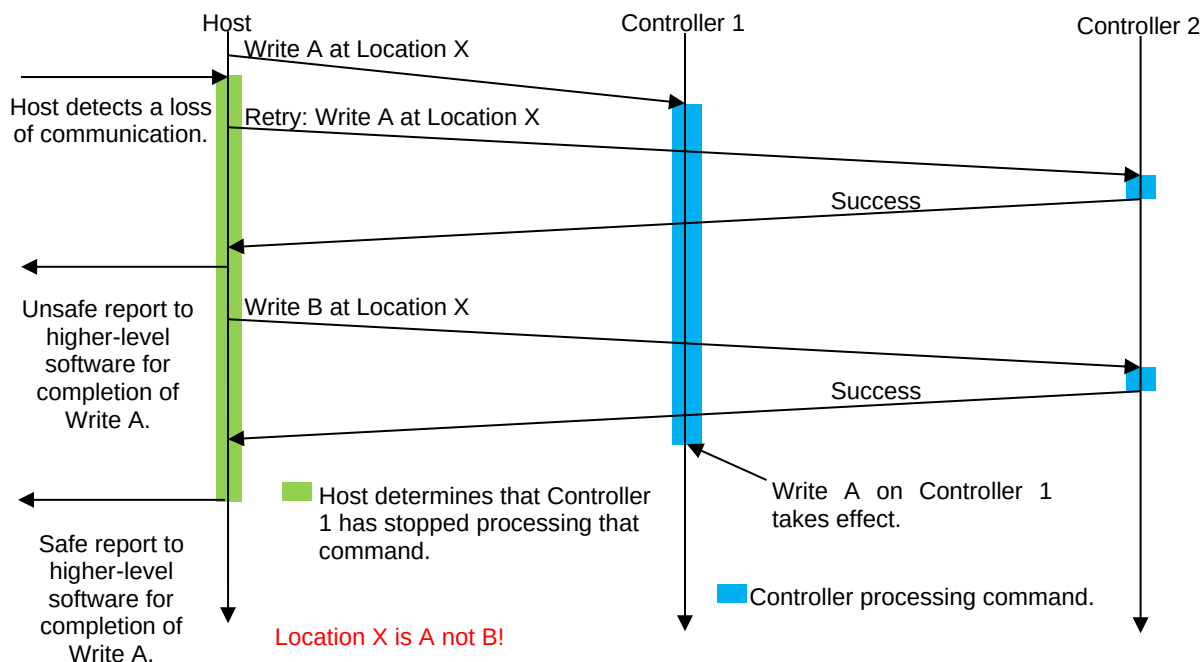
Bytes	Description										
00	Active Firmware Info (AFI): Specifies information about the active firmware revision.										
	<table><tr><th>Bits</th><th>Description</th></tr><tr><td>7</td><td>Reserved</td></tr><tr><td>6:4</td><td>Next Active Firmware Slot (NAFS): This field indicates the firmware slot that is going to be activated at the next Controller Level Reset. If this field is cleared to 0h, then the controller does not indicate the firmware slot that is going to be activated at the next Controller Level Reset.</td></tr><tr><td>3</td><td>Reserved</td></tr><tr><td>2:0</td><td>Current Active Firmware Slot (CAFS): This field indicates the firmware slot from which the actively running firmware revision was loaded</td></tr></table>	Bits	Description	7	Reserved	6:4	Next Active Firmware Slot (NAFS): This field indicates the firmware slot that is going to be activated at the next Controller Level Reset. If this field is cleared to 0h, then the controller does not indicate the firmware slot that is going to be activated at the next Controller Level Reset.	3	Reserved	2:0	Current Active Firmware Slot (CAFS): This field indicates the firmware slot from which the actively running firmware revision was loaded
	Bits	Description									
	7	Reserved									
	6:4	Next Active Firmware Slot (NAFS): This field indicates the firmware slot that is going to be activated at the next Controller Level Reset. If this field is cleared to 0h, then the controller does not indicate the firmware slot that is going to be activated at the next Controller Level Reset.									
3	Reserved										
2:0	Current Active Firmware Slot (CAFS): This field indicates the firmware slot from which the actively running firmware revision was loaded										
07:01	Reserved										
15:08	Firmware Revision for Slot 1 (FRS1): Contains the revision of the firmware downloaded to firmware slot 1. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
23:16	Firmware Revision for Slot 2 (FRS2): Contains the revision of the firmware downloaded to firmware slot 2. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
31:24	Firmware Revision for Slot 3 (FRS3): Contains the revision of the firmware downloaded to firmware slot 3. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
39:32	Firmware Revision for Slot 4 (FRS4): Contains the revision of the firmware downloaded to firmware slot 4. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
47:40	Firmware Revision for Slot 5 (FRS5): Contains the revision of the firmware downloaded to firmware slot 5. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
55:48	Firmware Revision for Slot 6 (FRS6): Contains the revision of the firmware downloaded to firmware slot 6. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
63:56	Firmware Revision for Slot 7 (FRS7): Contains the revision of the firmware downloaded to firmware slot 7. If no valid firmware revision is present or if this slot is unsupported, this field shall be cleared to 0h.										
511:64	Reserved										

5.2.12.1.5 Changed Attached Namespace List (Log Page Identifier 04h)

This log page is used to describe changes to attached namespaces for this controller, since the last time that this log page was read, that:

- a) have changed information in their Identify Namespace data structures (refer to section 1.5.50);
- b) were previously unattached to the controller and have since been attached to the controller;
- c) were previously attached to the controller and have since been detached from the controller; and
- d) were deleted.

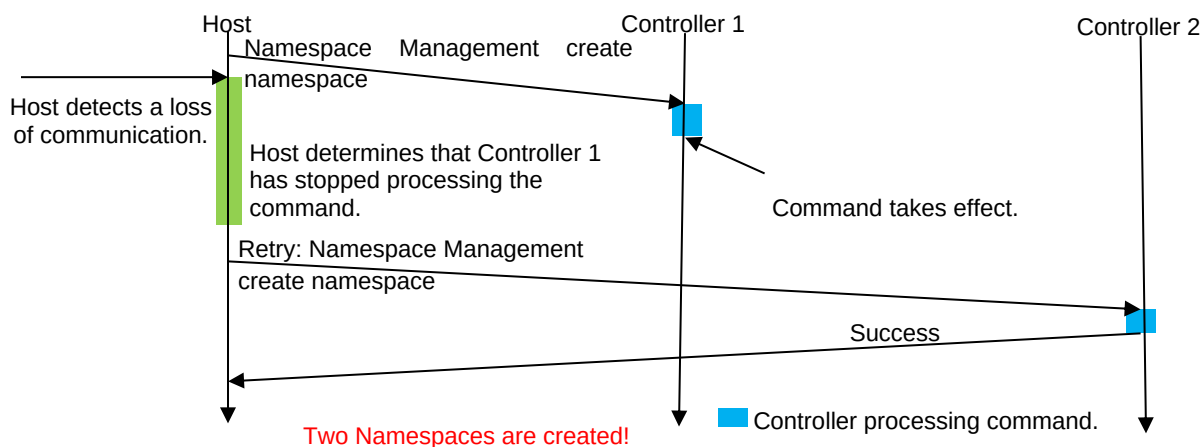
The log page contains a Namespace List with up to 1,024 entries. If more than 1,024 namespaces have changed attributes since the last time the log page was read, the first entry in the log page shall be set to FFFFFFFFh and the remainder of the list shall be zero filled.

Figure 817: Write after Write

B.6.2. Non-Idempotent Command

In the example shown in Figure 818, the host loses communication with Controller 1 and does not receive a response from Controller 1 for a Namespace Management command that creates a namespace (refer to section 5.2.21). The host ensures that no further controller processing of that command is possible (refer to section 9.6.2), and then retries that command on Controller 2, which creates a second namespace.

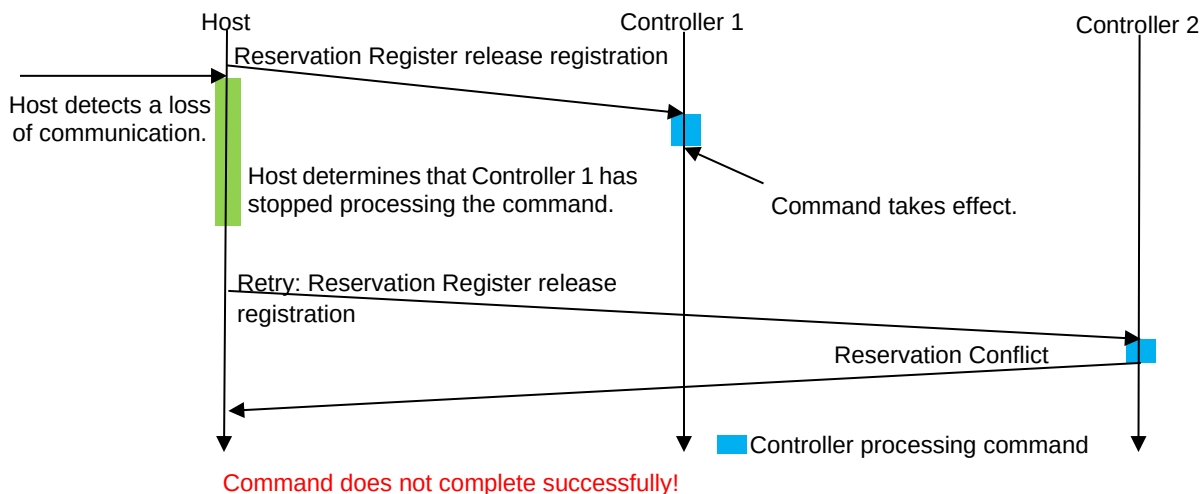
This example shows why higher-level software (e.g., an associated application, filesystem or database) should take steps to determine that a retry of a non-idempotent command does not cause unintended changes to NVM subsystem state (e.g., number of namespaces).

Figure 818: Non-Idempotent Command

B.6.3. Retried Command Does Not Succeed

In the example shown in Figure 819, the host loses communication with Controller 1 and does not receive a response to a Reservation Register command that unregisters the host (refer to section 7.6). The host ensures that no further controller processing of that command is possible (refer to section 9.6.2), and then retries that command on Controller 2. As a result of the original command unregistering the host, the host is no longer a registrant, and for that reason, the controller returns a status code of Reservation Conflict (refer to section 8.1.24.4).

Figure 819: Retried Command Does Not Succeed



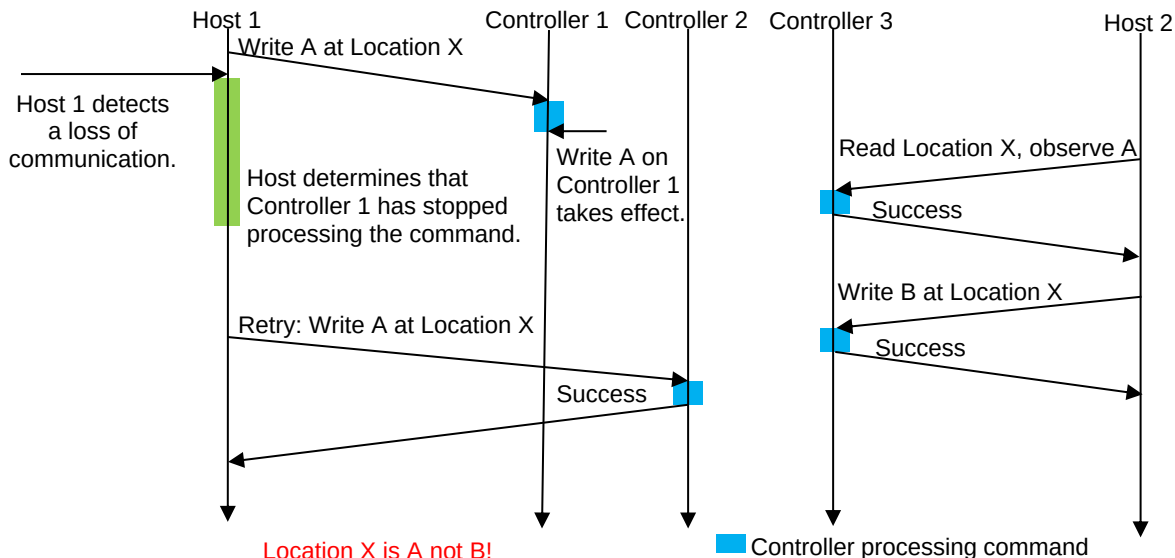
This example shows why an error status code is able to be returned if a non-idempotent command is retried after the original command has been processed. An analogous example is possible for the Compare and Write fused operation (refer to the Fused Operation section of the NVM Express NVM Command Set Specification) because that fused operation is not idempotent.

B.6.4. Retried Command Affects Another Host

In the example shown in Figure 820, two hosts use Location X in a namespace for coordination. Writing the value A to Location X indicates that step A in a processing sequence has been completed, and writing the value B indicates that step B in the processing sequence has been completed, where higher-level software requires that step B follow step A.

Host 1 indicates completion of step A by writing the value A to Location X, but loss of communication prevents Host 1 from receiving the completion of that command. Host 2 observes that step A is complete, quickly performs step B, and indicates completion of step B by writing the value B to Location X.

In the absence of receiving a completion for the original command, Host 1 retries writing the value A to Location X, overwriting the completion of step B reported by Host 2. This example shows that retry of commands that are able to affect the behavior of other hosts is problematic. In this example, higher-level software needs a mechanism to indicate that the writes to Location X are not safe to retry after a delay.

Figure 820: Retried Command Affects Another Host

This sort of higher-level software usage of ordinary NVMe commands (e.g., NVM Command Set Write commands) for coordination and synchronization among multiple hosts is strongly discouraged because retry of these commands after communication loss is problematic. Higher-level software should instead use mechanisms intended for coordination among multiple hosts. Two examples of such mechanisms are:

- Reservations (refer to section 8.1.24); and
- Compare and Write fused operations (refer to the Fused Operation section of the NVM Express NVM Command Set Specification).

In addition, command retries that modify NVM subsystem state (e.g., a Set Features command that modifies a feature that has any scope that is visible to other hosts as described in Figure 403) are able to affect the behavior of other hosts. Use of commands that modify NVM subsystem state for coordination and synchronization among multiple hosts is likewise strongly discouraged.

Annex C. Power Management and Consumption (Informative)

NVM Express power management capabilities allow the host to manage power for a controller. Power management includes both control and reporting mechanisms.

For information on transport power management (e.g., PCIe, RDMA), refer to the applicable NVM Express Transport specification.

The scope of NVM Express power management is the controller (refer to section 5.2.26.1.2).

NVM Express power management uses the following functionality:

- a) Features:
 - Power Management (refer to section 5.2.26.1.2 and section 8.1.18);
 - Autonomous Power State Transition (refer to section 5.2.26.1.6 and section 8.1.18.2);
 - Non-Operational Power State Configuration (refer to section 5.2.26.1.10 and section 8.1.18.1); and
 - Spinup Control (refer to section 5.2.26.1.18);
- b) NVM subsystem workloads (refer to 8.1.18.3); and
- c) Runtime D3 transitions (refer to section 8.1.18.4).

Controller thermal management may cause a transition to a lower power state, interacting with these Features:

- a) Temperature Threshold (refer to section 5.2.26.1.3);
- b) Host Controlled Thermal Management (refer to section 5.2.26.1.9 and section 8.1.18.5); and
- c) access to host memory buffer (refer to section 5.2.26.2.4) may be prohibited in non-operational power state.

NVM Express power management uses these reporting mechanisms:

- a) properties:
 - Controller Power Scope (CAP.CPS) (refer to Figure 36);
- b) fields in the Identify Controller data structure (refer to Figure 328):
 - RTD3 Resume Latency (RTD3R);
 - RTD3 Entry Latency (RTD3E);
 - Non-Operational Power State Permissive Mode;
 - Number of Power States Support (NPSS);
 - Autonomous Power State Transition Attributes (APSTA); and
 - Power State 0 Descriptor (PSD0) through Power State 31 Descriptor (PSD31) (refer to Figure 329);
- c) Features:
 - Power Management (refer to section 5.2.26.1.2);
 - Temperature Threshold (refer to section 5.2.26.1.3);
 - Autonomous Power State Transition (refer to section 5.2.26.1.6 and section 8.1.18.2);
 - Non-Operational Power State Configuration (refer to section 5.2.26.1.10 and section 8.1.18.1);
 - Host Controlled Thermal Management (refer to section 5.2.26.1.9 and section 8.1.18.5);
 - Host Memory Buffer (refer to section 5.2.26.2.4); and
 - Spinup Control (refer to section 5.2.26.1.18);and
- d) log pages:
 - SMART / Health Information log page fields (refer to section 5.2.12.1.3):
 - Thermal Management Temperature [1-2] Transition Count; and